Flume experiment

The handouts and information for this class are adapted from material developed by Dr Merren Jones.

Aim: The aim of this experiment is to explain what happens to a sand bed as a unidirectional water flow above it is increased in its capacity.

Objective: The objective of this experiment is to demonstrate the sequence of bedforms that appear in an open channel with a mobile, sand bed as the intensity of sediment transport increases.

In our experiment we will be running a unidirectional (one-way) water flow over a sand bed. The grain size of the sediment will be held constant (mean=255 μm; range 300-150 μm), as will the grain density (quartz), fluid density and viscosity (water @20°C).

Intended learning outcomes

At the end of this practical class you should be able to:

• use laboratory equipment to measure and record key variables during the experiment including flow depth, flow velocity, flow structure, bed slope, bedform migration rate

• illustrate your observations using interpreted, annotated diagrams which show both the physical observations (e.g. bedform and sedimentary structure) and processes of erosion, transport and deposition

• relate changes in the way in which particles move, the sediment transport rate and bedform created to changes in the flow strength

• work co-operatively in a research team to follow an experimental procedure to successfully complete the experiment in the time available. This will require you to evaluate your observations during the experiment and make adjustments to the instructions as required.

SAFETY NOTE: WATER AND ELECTRICITY ARE A LETHAL COMBINATION!!!!

Equipment Required

Flume; Vernier depth gage; Stop watch; Digital spirit level; Chinagraph pencil; Ruler Tape measure; Neutrally buoyant green balls; Red food colouring and pipette

Equipment Set Up

Sand of a uniform size between 0.1 and 0.3mm is placed in the flume to a uniform depth. The sand surface is tamped down so that it is uniformly packed, lightly compacted and as flat as possible.

The flow valve is fully closed.

The flume slope is set to zero.

You will work in research teams of between 4-6 members. You will need to divide the following control and measurement tasks among your team's members before you begin:

Job 1 • J.A.R.V.I.S. (in charge of the flume controls, including the Big Red Button, and generally keeping an eye on everything)

Job 2 • scribe

Job 3 • bed slope (using digital spirit level) • flow visualiser (using red dye and pipette)

Job 4 • flow depth (vernier depth gage)

Job 5 • flow velocity (using a stop watch and neutrally buoyant green ball)

Job 6 • bedform migration rate (using chinagraph pencil, stop watch and ruler)

Experimental Procedure

At each stage of the experiment be patient and allow the flow and bed to adjust to the new settings (i.e. reach some sort of equilibrium). At a minimum, allow the new bed state to be established through the middle working section of the flume.

Step 1: Ensure that the bed slope is set to a zero slope and open the flow control valve two turns. Turn on the pump, making sure that the pump is set to its lowest setting I, and allow the water to enter the flume.

Allow the tank to setting a couple of minutes. Under these conditions the bed should be immobile. Wait until steady flow along the length of the flume is achieved, and check that there is no bed motion.

To be successful, it is important to observe the condition of the bed closely and carefully during the rest of the experiment. Increase the slope a fraction, but not enough to exceed the threshold of bed material motion. To find this point you will need to decide how many grains need to be moving for the threshold of motion to be exceeded. Before the threshold is exceeded there is (almost) zero sediment transport and the bed is immobile. Under these conditions there are no bedforms and the configuration of the bed is that of a plane bed with no motion.

Actions required:

- Record the width of the flume, pump setting, whether the valve is open or closed, mean water depth and flume slope in Results Table 1.
- Calculate the surface water velocity determined using a stop watch and green ball.

Mean depth should be based on at least 10 point measurements of water depth made at random positions within the working reach of the flume (the central 50 cm) and well away from any entrance or exit effects. Note that the water depth is measured from the free surface to the top of the sand bed, not to the solid flume bed.

Questions: Is the flow structure laminar or turbulent?

• Visualise the flow structure by dropping some red dye into the flow- does it trace out a line or is it mixed?

Step 2: Increase the slope incrementally until there is a small but finite sediment load. Well done- you have initiated bedload transport!!!!!

Almost immediately small, triangular bedforms will appear. These are ripples. They are a few millimetres high and a centimetre or two long and are asymmetrical, with a gentle upstream slope and a steep downstream face.

Actions required:

- Observe the motion of grains making up the ripples.

- Observe the geometry of the water surface.
- Record the flow velocity, mean water depth and flume slope in Results Table 1.

Step 3: Increase the pump discharge to setting III.

The flow depth and rate of sediment movement will both be seen to increase immediately. If necessary, increase the slope incrementally until grains are seen to hop and bounce periodically, even though the predominant mode of motion is still by rolling.

Question: What is this motion by hopping or bouncing called?

Under this flow intensity, larger triangular bedforms called dunes become dominant. The size of dunes produced in a small flume is not much greater than that of the ripples. However, they are also longer and flatter topped, and disturb the free surface of the flow.

Once the dune bed has been established throughout the working section of the flume, you will see that the water surface is drawn down slightly over a dune crest, but is humped over the trough between dunes. The waves on the water surface are thus out of phase with the waves in the bed.

Actions required:

- Observe the process of sediment movement downstream, with rolling and saltation of grains up the stoss slope.
- Observe grains avalanching down the lee slope, and watch for the presence of separation bubbles on the lee slopes, causing upstream flow which sweeps grains back up the face.
- Observe the geometry of the water surface.
- Make a careful annotated sketch of these features, noting the height and geometry of the dune. Measure the characteristic dune heights and wavelengths.
- Record the migration rate of the dunes down flow using the chinagraph pencil, a stopwatch and ruler.
- Record the flow velocity, mean water depth and flume slope in Results Table 1.

Step 4: Open the flow valve fully so that you are producing maximum discharge.

The flow depth again increases, as does the intensity of sediment motion. Saltation will become more pronounced and the movement of some grains as suspended load maybe observed, as grains leap from the crest of one dune directly onto the stoss slope of the next without coming to rest in the trough.

Actions required:

- Note any changes in the dune height, length, and rate of progression downstream using the chinagraph pencil, a stopwatch and ruler.
- Record the flow velocity, mean water depth and flume slope in Results Table 1.

Step 5: Now make a substantial increase in slope

Actions required:

- Observe that the dunes lengthen and reduce in amplitude. The resulting long, low bedforms are called washed-out dunes. Note their shape and the presence (or lack of) troughs and scour holes.
- When the wash-out dunes occupy the whole working length, record the flow velocity, mean water depth and flume slope in Results Table 1.

NOTE: time is of the essence in the remaining parts of the experiment because the high rates of bed material transport associated with the next bedforms will overwhelm the capacity of the flume to recirculate sediment, resulting in rapid scour down to the solid flume bed. Measurements must be made as quickly as possible once each bedform regime has become established.

Step 6: Make a further substantial increase in flume slope.

Actions required:

- Observe the disappearance of the washed-out dunes and the establishment of a plane bed with motion.
- Note changes in the water depth and the appearance of the water surface, and the presence or absence of boils and surface turbulence.
- Note the appearance of the moving bedload.
- Record the flow velocity, mean water depth and flume slope in Results Table 1.

Step 7: Increase the flume slope by a small amount.

Actions required:

- Observe that the flat, glassy smooth water surface begins to deform into almost stationary undulations called standing waves.
- Observe the appearance of the flume bed, and note any relationship between the bed features and the flow surface.
- Record the mean water depth and flume slope in Results Table 1.

Step 8: Make a fairly substantial increase in flume slope to move the flow well into the upper regime.

Actions required:

- Observe the amplitude of the surface waves and the bedforms below them, and note the symmetry and regularity of the bedforms, which are known as antidunes.
- Observe the motion of grains as they are eroded from the downstream face of one antidune and deposited on the upstream face of the next antidune downstream. Motion of the antidune moves in the upstream direction, against the movement of the water flow and the grain motion.
- This could be verified by tracing the outline of an antidune on the flume wall using a chinagraph pencil and observing its movement through time. It must be recognised that the water and the grains move downstream at all times, even though the bedform moves upstream.
- Record the flume slope in Results Table 1